Topic 10 — Climate Change (जलवायु परिवर्तन)

Covers syllabus (click to expand)

Global Warming (भूमंडलीय तापन) | Greenhouse Effect | Greenhouse Gases (GHGs) | Carbon Footprint | Carbon Sink (कार्बन सिंक) | Carbon Sequestration (कार्बन पृथक्करण) | Climate Change Impacts | Climate Adaptation (जलवायु अनुकूलन) | Climate Mitigation (जलवायु शमन)

Sources baked in: IPCC AR6, NCERT Geography/Biology, India NDC 2022, NAPCC 2008, MoEFCC, IMD, WMO GHG Bulletin shares, UPPCS/UPSC PYQs 2018–2025 **Weight:** ★★★ High — GHG traps, adaptation vs mitigation, treaty years, India targets **Last verified:** September 2026

Current Affairs (this topic)

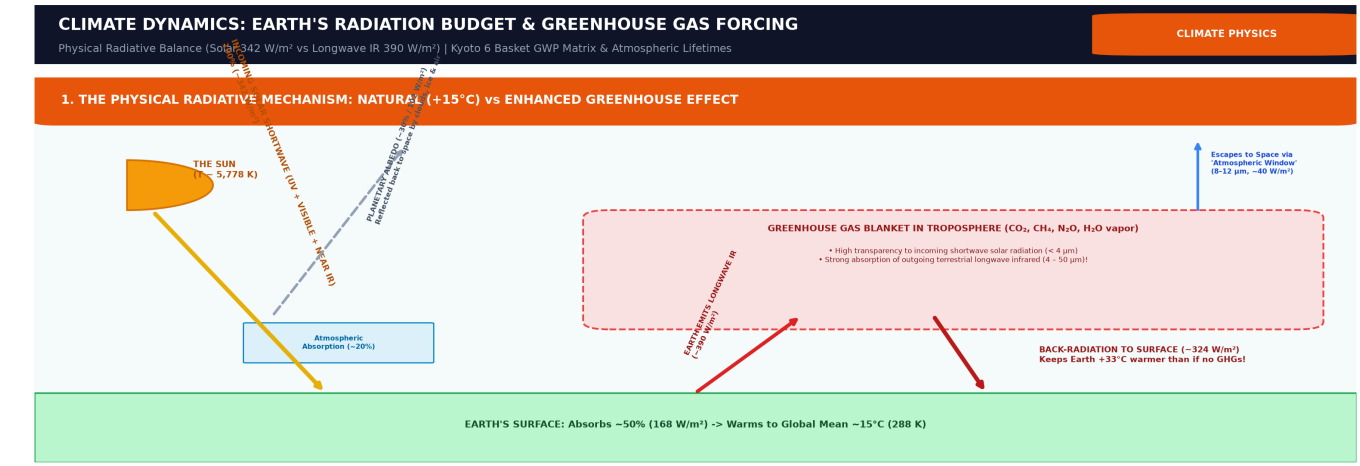
- Keep India NDC / Net Zero 2070 / LiFE as living climate CA; treaty years stay in the ladder below.
- Do not put **Montreal** into a climate-mitigation answer slot when Kyoto (क्योटो) / Paris are present.

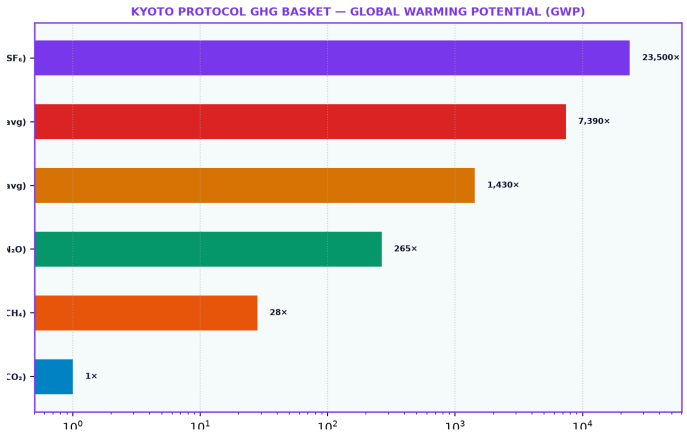
10.1 Global Warming

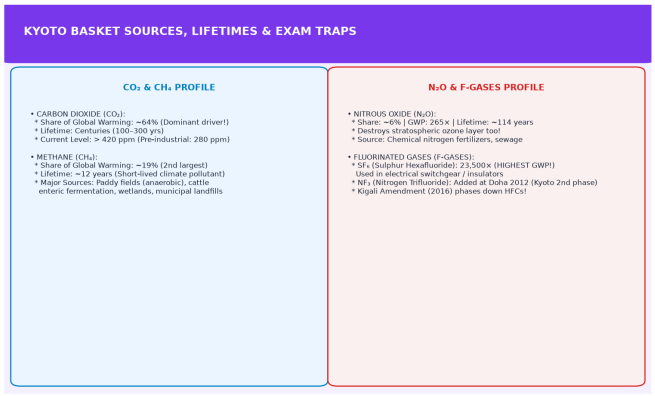
Identity: Temperature rise since industrial era | IPCC 1.1°C | Anthropogenic dominant

- **Global warming** is the long-term increase in Earth's average near-surface temperature.
- **Climate change** is broader and includes shifts in rainfall, monsoon timing, cyclones, and extremes.
- Since the industrial era, atmospheric **CO_2** has risen from about **280 ppm to over 420 ppm**.
- **IPCC AR6** estimates (प्राकलन समिति) warming of about **1.1°C** above the **1850–1900** pre-industrial baseline.
- **Human activities** — fossil fuels, deforestation, industry, agriculture — are the **dominant cause** of recent warming.
- **India** has warmed about **0.7°C** since 1901 according to **IMD** data.
- **Oceans absorb about 90%** of excess heat, causing thermal expansion and marine stress.
- **Arctic warming** is about **2–3 times faster** than the global average.
- A single cold day is **weather**. Multi-decade warming trend is **climate**.

Logic: "Warming is entirely natural" is **false**. IPCC attributes recent trend mainly to human activity.







10.2 Greenhouse Effect

Identity: Fourier 1820s | Shortwave in · longwave out | Natural vs enhanced | ≠ ozone depletion (ओजोन क्षरण)

- The **greenhouse effect** is the process by which atmospheric gases absorb and re-emit **outgoing infrared radiation**, warming the surface and lower atmosphere.
- **Joseph Fourier (1820s)** postulated the greenhouse-effect concept.
- The Sun emits **high-energy shortwave** radiation (UV, visible, near-IR). Harmful **UV** is largely absorbed by the **ozone layer**; visible and IR reach the surface as heat.
- Earth, being cooler, re-emits energy as **longwave IR**. Greenhouse gases absorb much of that longwave and re-radiate it in all directions.
- The **natural greenhouse effect** keeps Earth's average temperature near **+15°C**.
- Without it, Earth's average would be about **−18°C** and largely uninhabitable.
- The **enhanced greenhouse effect** comes from human-caused GHG buildup faster than natural sinks can absorb — this drives **global warming**.
- **Water vapour** is the most abundant natural GHG (about **half** of the natural greenhouse effect in NASA-style framing) but acts mainly as a **feedback**, not a primary emitted driver like fossil CO_2.
- The greenhouse effect is **not ozone depletion**. Ozone blocks UV; GHGs trap IR.

> **Logic:** Natural greenhouse effect is **essential**. The problem is the **enhanced** anthropogenic effect. Fourier coined the concept — do not invent modern IPCC authors for that stem.

10.3 Greenhouse Gases (GHGs)

Identity: Kyoto basket | WMO share order | Argon/H_2/propane not GHG | Direct vs indirect

- **Greenhouse gases** absorb and emit infrared radiation in the atmosphere.
- **CO_2** is the main anthropogenic driver by concentration and total forcing. IPCC-style forcing share is often ~**76%**; older WMO-bulletin style relative shares used in many Indian papers are **CO_2 ~64%, CH_4 ~19%, N_2O ~6%, CFCs + others ~11%**.
- Decreasing contribution order commonly tested: **CO_2 > CH_4 > CFCs > N_2O**.
- **CO_2** has **GWP = 1**. Pre-industrial ~**278–280 ppm (~0.03%)**; ~**315 ppm** around 1950s; now **415–420+ ppm**.
- **Methane (CH_4)** — also called **marsh gas** — has **GWP around 28–36** (100-yr). Sources: paddies, cattle, coal mines, landfills, termites; natural **wetlands** dominate natural CH_4 (~**75–76%**).
- **Rice** is a major anthropogenic source of **both methane and nitrous oxide** (anaerobic soils + nitrogen fertiliser).
- **Nitrous oxide (N_2O)** has **GWP around 265** and links strongly to synthetic fertilisers.
- The **Kyoto Protocol** covers **CO_2, CH_4, N_2O, HFCs, PFCs, and SF_6** (later inventories also track **NF_3**).
- **Not greenhouse gases: argon, N_2, O_2, hydrogen, propane**. **Nitrogen oxide (NO_x)** is **not** listed among the classic naturally occurring GHGs in UPPCS-style keys.
- **SO_2 and oxides of nitrogen** do **not** contribute to global warming **directly** (acid rain (अम्ल वर्षा) / air pollution / indirect chemistry roles).
- **IPCC direct GHGs** include CO_2, CH_4, N_2O, halocarbons, SF_6, NF_3. **Indirect** agents include **NO_x, CO, NMVOCs, SO_2**.
- **Carbon fertilization** means increased plant growth from higher atmospheric CO_2 — distinct from warming or ocean acidification.
- Oceans remove much atmospheric CO_2 via **phytoplankton photosynthesis** (biological pump) — destroying phytoplankton harms the ocean carbon sink and marine food webs, not seawater density.
- **Black carbon (soot)** is a climate forcer but **not a GHG**.
- **India is the 3rd largest total GHG emitter** (**China** #1); **low per capita**. **Bhutan** is often described as **carbon-negative**.

GHG / group	Rough relative share (WMO-style stems)	GWP (100-yr)
CO_2	~64%	1
CH_4	~19%	28–36
N_2O	~6%	265
CFCs + others	~11%	Very high (F-gases)

> **Logic:** Highest **GWP per molecule** ≠ largest total warming. **CO_2 dominates by volume**. Water vapour can win “most abundant natural GHG / largest share of natural effect” stems; CO_2 wins “main component / highest contribution to anthropogenic warming” stems.

10.4 Carbon Footprint

Identity: tCO_2e | Scope 1/2/3 | India low per capita, high total

- A **carbon footprint** is the total GHG emissions caused by a person, product, organisation, or country.
- It is measured in **tonnes of CO_2 equivalent (tCO_2e)** using GWP conversion.
- **Scope 1** covers **direct** emissions from owned sources such as factory (फैक्टरी) boilers.
- **Scope 2** covers **indirect** emissions from purchased electricity or heat.
- **Scope 3** covers other indirect emissions such as supply chains, travel, and product disposal.
- **India's per capita footprint** is about **2 tCO_2e**, below the global average of about **4.7**.
- The **energy/electricity sector** is India's largest emissions source.
- **LiFE Mission (2022)** promotes lifestyle changes to reduce individual footprints.

> **Logic:** India has low **per capita** footprint but high **total** emissions due to population (जनसंख्या) size.

10.5 Carbon Sink

Identity: Net CO_2 absorption | Forests, oceans, soil | Opposite = source

- A **carbon sink** absorbs more CO_2 than it releases over time.
- **Forests** store carbon in biomass and soil through photosynthesis.
- **Oceans** absorb about **25% of anthropogenic CO_2** through dissolution (विघटन) and the biological pump.
- **Soils and wetlands** store organic carbon. Degraded land can become a **carbon source**.
- **Mangroves** are **blue carbon** sinks with very high sequestration rates per hectare.
- **Deforestation** converts a sink into a **source** by releasing stored carbon and removing uptake capacity.
- India's NDC targets an additional **2.5–3 GtCO_2e forest carbon sink** by 2030.

> **Logic:** A coal plant is a **source**. A growing forest is a **sink**.

10.6 Carbon Sequestration

Identity: Long-term carbon storage | Biological + geological | Farm practices | CCS (सुरक्षा कैबिनेट समिति) sites

- **Carbon sequestration** is the process of capturing and storing atmospheric carbon for long periods.
- **Biological sequestration** uses afforestation, soil carbon build-up, and mangrove restoration.
- Farm practices that help soil carbon storage include **contour bunding**, **relay cropping**, and **zero tillage**.

- **Geological sequestration / CCS** can use abandoned coal seams, depleted oil and gas reservoirs, and deep saline (लवणीय) formations; other discussed methods include crushed basalt on farmland and ocean alkalinity enhancement.
- **Afforestation and reforestation** are India's main practical sequestration tools under **Green India Mission**.
- **Blue carbon** is carbon captured by oceans and coastal ecosystems (mangroves, seagrasses, salt marshes).
- **CAMPA** and the **Green Credit Programme (2023)** finance ecosystem (पारिस्थितिकी तंत्र) restoration in India.
- Stored carbon can be **re-released** by fire or deforestation, creating permanence risk.

> **Logic:** Sequestration is the **process**. Sink is the **reservoir**. Blue carbon ≠ forest biomass carbon label.

10.7 Climate Change Impacts

Identity: Heat, glaciers, monsoon extremes, agriculture, coral bleaching | Arctic first

- **Heat waves** are increasing in frequency across India, especially in **northwest India and UP**.
- **Himalayan glaciers** such as Gangotri (गंगोत्री) are retreating, affecting lean-season Ganga (गंगा) flow.
- **Monsoon** shows more **extreme rainfall bursts** and longer dry spells rather than uniform change.
- **Bundelkhand** faces recurring **drought** and water scarcity.
- **Eastern UP** along the Ganga faces **flood risk** from intense monsoon events.
- **Wheat** in UP plains is sensitive to **March heat spikes** during the rabi (रबी) season.
- **Sea-level rise** threatens coastal India and increases delta (डेल्टा) salinity pressures.
- **Coral bleaching** occurs when sea surface temperature rises about **1–2°C above normal** for sustained periods.
- Impacts of warming commonly listed together: rise in sea level, melting glaciers, spread of diseases, coral bleaching, stronger cyclones/storms/hurricanes.
- The **Arctic and Greenland ice sheet** are among the most fragile systems hit first by warming.
- Rough IPCC risk framing: above about **+2°C** pre-industrial → widespread coral mortality; above about **+3°C** → terrestrial biosphere tends toward a **net carbon source** (wetlands do not all permanently disappear; cereal farming does not become impossible everywhere).
- Climate stress on agriculture and food security can raise **social tension** through resource pressure and food insecurity.
- **Milankovitch** astronomical climate theory covers **eccentricity, obliquity, and precession — not** solar irradiance as a listed element of that set.
- **Ice cores** provide the classic **cryogenic** record of past climate and atmospheric composition.
- Primary causes of recent climate change: fossil-fuel burning, oil-driven automobile growth, deforestation — **not** increasing solar flares as a major driver.

> **Logic:** — Andaman (अंडमान) coral bleaching linked to **+2°C Indian Ocean warming**. Do not mark "all wetlands vanish" or "no cereals anywhere" as automatic +3°C outcomes.

PYQ — UPPCS Prelims 2024, Q19

Coral bleaching in Andaman & Nicobar (निकोबार) caused by:

A. Coastal industry

B. +2°C Indian Ocean warming

C. Heavy pollution

D. –4°C cooling

Show answer

Ans: B

10.8 Climate Adaptation

Identity: Build resilience | NAPCC 2008 | 8 missions | ≠ mitigation

- **Climate adaptation** means adjusting to actual or expected climate impacts to reduce harm.
- **Mitigation** reduces emissions. **Adaptation** copes with impacts already occurring or already committed in the climate system.
- **NAPCC (National Action Plan on Climate Change, 2008)** is India's master climate framework with **8 national missions**.
- Adaptation-focused missions include **National Water Mission**, **Sustainable Agriculture Mission**, and **Himalayan Ecosystem Mission**.
- **National Water Mission** goals include a public water database and climate-impact assessment, citizen/state conservation action, focus on over-exploited areas, **~20% water-use efficiency gain**, and basin-level IWRM. Urban (नगरीय) wastewater recycling fits; nationwide Himalaya (हिमालय)–peninsula river linking and free borewell reimbursement do **not**.
- **Early warning systems** from IMD and CWC (केंद्रीय जल आयोग) for heat, floods, and cyclones are adaptation tools.
- **Climate-resilient crops**, micro-irrigation, and **PM Fasal Bima Yojana** support farm adaptation.
- **Bundelkhand water harvesting** and **Lucknow heat action plans** are UP adaptation examples.

#	NAPCC Mission
1	National Solar Mission
2	Enhanced Energy Efficiency
3	Sustainable Habitat (वास स्थान)
4	National Water Mission

#	NAPCC Mission
5	Sustaining Himalayan Ecosystem
6	Green India
7	Sustainable Agriculture
8	Strategic Knowledge for Climate Change

- **Atomic / nuclear power is not** one of the eight NAPCC missions.
- Climate-smart village approaches linked to **CCAFS / CGIAR** (HQ France; **ICRISAT** is a CGIAR centre) sit in the adaptation–agriculture research lane.

> **Logic:** Building sea walls = **adaptation**. Closing coal plants = **mitigation**. Nuclear ≠ listed NAPCC mission.

10.9 Climate Mitigation

Identity: Cut emissions + sinks | UNFCCC/Kyoto/Paris | CDM · GCF | India NDC | LiFE

Treaty ladder — climate vs ozone (read the stem)

Ladder	Instruments	What it controls	Classic trap
Climate	UNFCCC (1992) (जलवायु परिवर्तन संधि) → **Kyoto** (क्योटो) (1997) → **Paris** (2015)	GHG mitigation / NDCs	Montreal is **not** climate mitigation
Ozone	Vienna (1985) → **Montreal** (1987) → **Kigali** (HFCs, 2016)	ODS / HFC phase-down	Kyoto is **not** ozone

- **UNFCCC** (यूएनएफसीसीसी) is the parent climate convention. **Kyoto** gave binding Annex-I cuts and CDM. **Paris** uses voluntary **NDCs** for all parties.
- **CBDR** (common but differentiated responsibilities) is the climate equity line — do not paste it onto Montreal automatically.
- **Montreal** protects the ozone layer; **Kigali** amends Montreal for HFCs (also climate-relevant, but still ozone-family architecture).
- **Climate mitigation** means reducing GHG sources or enhancing sinks to limit future warming.
- **UNFCCC (1992, Rio)** entered into force **21 March 1994**; it frames greenhouse-gas mitigation.
- **Kyoto Protocol (1997; force 16 February 2005)** was the first binding emission-cut treaty for **developed (Annex I) countries**.
- Kyoto's **Clean Development Mechanism (CDM)** lets Annex-I parties fund projects in developing countries and earn **CERs** (1 CER ≈ **1 tCO_2**). Carbon credits are **market-traded**, not fixed by UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) price.

- **Paris Agreement (पेरिस समझौता) (2015)** (पेरिस समझौता) uses voluntary **NDCs** to limit warming well below **2°C** and pursue **1.5°C**. The "1000 billion/year" paraphrase is wrong — finance language was about mobilising **100 billion** class climate finance.
- **Green Climate Fund (GCF)** (जीसीएफ) was established at **Cancun COP-16 (2010)** under UNFCCC to help developing countries with adaptation and mitigation — **not** founded at Durban and **not** under UNEP/OECD/ADB/World Bank as its founding aegis.
- **India's updated NDC (2022)** targets **45% lower emissions intensity of GDP by 2030** (vs 2005).
- India targets **50% cumulative electric installed capacity from non-fossil sources by 2030** (not a "60% of total energy from renewables" paraphrase).
- India pledged **Net Zero by 2070** at **COP-26 Glasgow (2021)**. **Net zero** means emissions are **balanced by absorption/removal** — not that every emission falls to absolute zero.
- **LiFE (Lifestyle for Environment (जीवनशैली पर्यावरण))** was launched in **June 2022**. The idea was proposed at **COP-26**, not COP-25.
- **Earth Hour** is a **WWF** awareness event (usually last Saturday of March, lights off ~1 hour) — not UNEP/UNESCO.
- **Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (1987)** (मॉन्ट्रियल प्रोटोकॉल) controls **ozone-depleting substances**, not climate change.
- **GHG Protocol** (WRI + WBCSD) is an accounting tool for governments and business — not a UN treaty.

> **Logic:** Climate mitigation treaty stem → **Kyoto**. Montreal = ozone. GCF = Cancun 2010. Net zero ≠ absolute zero emissions.

PYQ — UPPCS Prelims 2024, Q35

Which agreement relates to climate change mitigation?

A. Montreal Protocol

B. Ramsar (रामसर)

C. Kyoto Protocol

D. CITES

Show answer

Ans: C

PYQ — UPPCS Prelims 2025, Q139

LiFE: (1) Launched June 2022. (2) Idea proposed at COP-25 Madrid.

A. Only 2

B. Neither

C. Both

D. Only 1

Show answer

Ans: D — Idea at **COP-26 Glasgow**.

Treaty	Year	Focus
UNFCCC	1992	Framework
Kyoto Protocol	1997	Binding cuts (developed countries)
Paris Agreement	2015	NDC pledges
Montreal Protocol	1987	**Ozone** (not climate)